

MID SEM – 2 (PHIC 103)

Max. Marks: 15

B.Tech (1st year: 2nd Semester)

Time Allowed: 50 min.

NOTE: Attempt all the questions. Extra ANSWER SHEETS will NOT be supplied.

1. Define mobility of the charge carrier in a typical semiconductor. Discuss the mechanisms of lattice and impurity scattering and describe how it affects the mobility of the charge carrier. [2 + 3]
2. Explain about the diffusion current density in semiconductors? An n-type semiconductor sample with $N_d = 10^{16} \text{ cm}^{-3}$ is under $E = 10 \text{ V/cm}$. Determine the resistivity of the sample at room temperature if $\mu_n = 8500 \text{ cm}^2/\text{V-s}$ and $\mu_p = 400 \text{ cm}^2/\text{V-s}$. Which experiment is suitable for measuring this resistivity? [2 + 2 + 1]
3. State the conditions for Rectifying and Ohmic contacts in a metal-semiconductor junction using diagrams. Show how I-V Characteristics look for both the cases. Sketch a neat and well labelled band diagram for metal-n type semiconductor rectifying junction under equilibrium. (No description needed) [2 + 1 + 2]